

SATNA, India

WMO: 425710

Lat: 24.567N

Lon: 80.833E

Elev: 1040

StdP: 14.15

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 45.5	(c) 47.9	(d) 29.9	(e) 24.9	(f) 83.0	(g) 33.4	(h) 29.1	(i) 78.7	(j) 5.5	(k) 66.8	(l) 5.2	(m) 67.5	(n) 0.1	(o) 320	(p) 0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.4	109.7	71.5	107.8	71.7	105.6	71.5	81.8	90.0	81.0	88.7	80.6	87.9	3.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	162.0	85.1	79.4	158.5	84.4	78.9	155.9	83.9	46.8	90.4	46.0	89.0	45.4	88.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.4	(c) 4.9	(e) 41.6	(f) 113.0	(g) 2.2	(h) 1.8	(i) 40.0	(j) 114.3	(k) 38.8	(l) 115.3	(m) 37.5	(n) 116.3	(o) 35.9	(p) 117.6		
(a) 6.7	(b) 5.4	(c) 4.9	(e) 39.4	(f) 84.3	(g) 2.2	(h) 1.3	(i) 37.8	(j) 85.3	(k) 36.5	(l) 86.0	(m) 35.3	(n) 86.8	(o) 33.7	(p) 87.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.3	61.8	68.6	78.1	88.7	94.4	92.3	84.3	83.0	83.0	79.8	72.1	64.4		
	DBStd	10.96	4.69	4.94	5.42	4.30	3.63	5.72	3.59	2.62	2.55	3.59	3.72	4.18		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	200	120	18	0	0	0	0	0	0	0	0	1	61		
	CDD50	10675	367	522	870	1161	1375	1270	1063	1023	991	924	662	447		
	CDD65	5402	23	120	406	711	910	820	598	558	541	459	212	44		
	CDH74	71434	270	1165	4692	10826	15216	12955	7185	6261	6048	4618	1813	385		
(13)	CDH80	39804	36	365	2512	7059	10816	8731	3051	2228	2360	2066	540	40		
(14)	Wind	WSAvg	1.1	0.7	0.9	1.1	1.4	1.7	2.0	1.6	1.4	1.2	0.8	0.4		
(15) Precipitation	PrecAvg	42.6	0.7	0.8	0.6	0.3	0.6	4.9	12.2	13.0	7.1	1.9	0.3	0.2		
	PrecMax	66.7	3.3	4.1	1.7	1.9	3.5	14.3	34.1	21.9	30.3	9.2	2.7	2.9		
	PrecMin	24.5	0.2	0.0	0.0	0.0	0.0	0.6	5.2	5.9	0.2	0.3	0.0	0.0		
	PrecStd	10.4	0.7	1.1	0.6	0.5	0.8	3.9	6.2	5.3	6.2	2.2	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	91.6	102.2	109.0	112.2	113.0	101.5	94.8	95.3	96.2	89.4	83.9		
		MCWB	63.4	64.9	66.8	68.9	70.5	75.3	78.0	78.8	76.6	69.9	67.3	63.9		
	2%	DB	80.3	88.3	98.5	106.7	110.3	109.7	96.4	91.8	93.2	93.7	87.1	80.6		
		MCWB	61.6	64.5	65.2	69.1	70.8	73.6	79.4	79.8	77.1	71.4	66.9	62.1		
	5%	DB	76.9	84.6	95.4	104.9	108.5	107.1	93.4	90.1	91.6	91.6	85.4	78.2		
		MCWB	59.7	62.6	65.0	68.5	71.1	73.8	79.4	79.9	77.2	71.6	66.1	61.0		
	10%	DB	73.4	81.4	92.5	102.7	106.5	104.4	90.8	88.5	89.9	89.6	83.4	75.9		
		MCWB	58.0	61.8	64.9	67.8	71.3	74.8	79.6	79.6	77.4	71.0	64.9	59.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	69.6	72.5	78.7	81.3	83.2	82.5	82.5	81.7	79.6	74.7	67.3		
		MCDB	79.1	83.3	90.2	104.1	100.7	95.7	90.5	88.8	88.1	87.3	84.7	78.7		
	2%	WB	63.6	67.1	70.1	73.5	78.5	81.4	81.7	81.5	80.7	78.3	72.1	65.3		
		MCDB	75.0	81.7	87.6	99.0	98.3	92.7	89.1	87.5	87.4	85.7	81.3	76.5		
	5%	WB	61.9	65.3	68.4	71.6	76.8	80.3	81.0	80.8	79.9	77.1	69.4	63.3		
		MCDB	71.7	79.1	86.4	95.9	96.8	91.7	88.2	86.8	86.5	84.1	79.4	73.8		
	10%	WB	60.2	63.6	67.0	70.1	75.4	79.4	80.5	80.2	79.2	75.9	67.4	61.6		
		MCDB	69.5	76.5	84.7	94.7	96.8	90.7	87.4	85.9	85.7	82.8	78.1	72.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	23.9	25.4	26.1	23.4	17.5	10.6	9.6	12.2	19.9	24.4	24.2		
		MCDBR	27.8	27.6	29.2	27.9	25.5	22.3	15.1	12.3	15.7	22.1	24.8	26.0		
		MCWBR	11.7	9.5	8.2	8.5	7.2	6.1	4.7	4.0	4.4	5.7	8.5	10.3		
	5% WB	MCDBR	21.1	23.1	24.4	25.1	21.4	16.1	11.5	10.7	11.9	13.9	19.6	21.7		
		MCWBR	10.0	9.5	8.4	9.5	8.2	5.4	4.7	4.3	4.5	4.7	9.0	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.485	0.472	0.490	0.554	0.628	0.680	0.633	0.586	0.559	0.572	0.579	0.532		
	taud		1.909	2.001	1.979	1.854	1.737	1.683	1.795	1.932	1.969	1.849	1.746	1.822		
	Ebn at Noon		239	255	258	244	226	213	223	233	236	224	209	217		
	Edh at Noon		55	54	57	66	75	78	70	61	57	62	64	58		
(44) All-Sky Solar Radiation	RadAvg		1256	1583	1945	2187	2196	1764	1391	1384	1524	1571	1351	1217		
	RadStd		120	95	89	73	64	161	146	115	164	102	78	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page